

WEEK 7

PROJECTS

- Allen Projects: mouse / human brain atlas - (Seattle) Now: mouse visual system
- Janelia farm - ^{Wash.} DC anatomy & physiology
- EU Human Brain Project - EPFL - Lausanne !!!
- President Obama's "Brain Activity Map Initiative" → Record many spheres in brains...
- Hagai Bergman.

Blue Brain Project

Database the brain

from genes to behaviour → collect all data.

Simulation based on the data → get understanding

- Henry Markram (Lausanne)
- Juan Luis Medina (Madrid)
- Idan Segev (Jerusalem)

$$1 \text{ mm}^2 = 30.000 \text{ cells}$$

100 ^{million} ~~000~~ connections

200 000 million ^{per} neocortex

Bert Sakmann -

Clog
Reed → Column - how it is organized
discovered that we have an entire column $(2n \times 2)^{n \times V1}$ responding to particular inputs

electrical
cell types

cell type anatomically

+

cell type electrically → write the equation for firing properties.

extended Hodgkin-Huxley

+

Active cable

↓

Multiple Objective Optimization (MOO)

How do you relate particular activity with particular behaviour?

In Amsterdam → living tissue (cortex)

Nowadays we don't have enough tools...

HUMAN BRAIN PROJECT

Henry Markram

Platform for understanding the brain & developing new technologies

Goals:
└ new understanding of the brain
└ new ITs for diseases.
└ new brain-like computing technologies

Neuroinformatics / Brain Simulation / Medical Info.

High Performance Computing / Neuroanalogic Computing / Neurobotics